

Structural and Theoretical Enhancements to the ROSE Federated Learning Algorithm: A Comparative Paradigm Against the SHAPEFL Framework

Introduction to the Distributed Optimization Landscape

The paradigm of Federated Learning has fundamentally restructured the approach to distributed machine learning and large-scale optimization, enabling the collaborative training of global models across decentralized nodes without compromising the privacy of raw local data.¹ By transferring the computation to the edge and only exchanging model parameters or gradients, this approach addresses critical concerns regarding data sovereignty, regulatory compliance, and raw data transmission costs. However, the transition from theoretical models to real-world deployment across heterogeneous networks has exposed critical vulnerabilities and bottlenecks in both cross-device and cross-silo architectures. These bottlenecks primarily manifest as severe communication overhead due to high-dimensional parameter exchanges, latency induced by synchronous aggregation, and susceptibility to adversarial perturbations, such as Byzantine attacks and membership inference attacks.² Furthermore, statistical heterogeneity—commonly referred to as non-Independent and Identically Distributed (non-IID) data—significantly degrades convergence rates, as the local objectives diverge from the global empirical risk minimization objective.²

Within this complex landscape, two distinct methodologies have emerged to address disparate facets of the distributed optimization challenge. The ROSE (Robust Optimization for Secure Environments) implementation prioritizes security and privacy enhancement. It utilizes a Model Update Based (MUB) federated averaging algorithm combined with individual client model initialization to defend against Byzantine attacks, such as additive noise and sign-flipping, as well as membership inference attacks.² The architectural philosophy of ROSE dictates that security must not be compromised, even in untrusted edge networks. Conversely, the SHAPEFL (SHaping dAta distRibution at Edge) framework introduces a Communication-Efficient Hierarchical Federated Learning architecture. SHAPEFL explicitly mitigates communication costs by interposing edge aggregators between the clients and the central cloud, actively shaping the data distribution at the edge to reduce the required number of global communication rounds.¹

While both frameworks demonstrate significant advancements in their respective domains, they exhibit mutually exclusive limitations. SHAPEFL achieves communication efficiency but incurs prohibitive computational costs due to its algorithmic reliance on cosine distance for data diversity measurement and requires offline pre-training, rendering it rigid in dynamic, real-world environments.⁷ The ROSE implementation achieves high adversarial resilience but suffers from the classical communication bottlenecks of flat architectures and experiences substantially degraded convergence speeds under highly non-IID data distributions.²

This research report presents an exhaustive comparative analysis of the structural and theoretical underpinnings of both the ROSE implementation and the SHAPEFL framework. By identifying the algorithmic and structural bottlenecks in both approaches derived from an analysis of their base paper logic and parser implementations, a novel, integrated architectural enhancement to the ROSE algorithm is proposed. This enhanced framework, serving as the foundation for an advanced academic thesis, aims to supersede the performance metrics of SHAPEFL in convergence speed, communication efficiency, and model accuracy, while retaining the uncompromising Byzantine fault tolerance inherent to the ROSE methodology.

Architectural and Theoretical Analysis of the SHAPEFL Framework

The SHAPEFL framework was engineered specifically to address the Communication Cost Minimization (CCM) problem inherent in traditional single-tier federated networks. When data across distributed computing nodes is highly imbalanced or non-IID, traditional aggregation algorithms like Federated Averaging require a significantly higher number of communication iterations to achieve model convergence.¹ SHAPEFL addresses this statistical inefficiency by establishing a hierarchical topology consisting of distributed computing nodes, intermediate edge aggregators, and a central cloud server.

The Communication Cost Minimization Problem

The core theoretical contribution of the SHAPEFL architecture is the mathematical formalization of the CCM problem. The framework defines the variable κ as the total number of cloud aggregations required to reach a predetermined target learning accuracy.¹ The total communication cost between the distributed computing nodes, denoted as the set N , and the edge aggregators, denoted as the set N_e , is represented by the following objective function:

$$C_{ne}(Y, \kappa) = \sum_{n \in N} \sum_{e \in N_e} \kappa c_{yme} c_{ne}$$

Within this formulation, Y represents the node-edge association matrix, c_{yme} is the binary

association variable indicating whether node n is connected to edge e , and c_{ne} represents the communication cost per round between node n and edge e .⁶

To optimize this objective function and make the problem computationally tractable, SHAPEFL divides the original optimization into two distinct sub-problems. The first sub-problem focuses on minimizing the per-round communication cost by establishing the optimal physical and logical associations between clients and edge servers, effectively solving for the optimal matrix Y .¹⁰ The second sub-problem focuses on maximizing data distribution diversity at the edge layer. By ensuring that the aggregated data at each edge server is as diverse and IID-like as possible, the framework empirically reduces κ , the total number of global cloud aggregations required for the overarching model to converge.⁶

Algorithmic Bottlenecks in the SHAPEFL Parser Logic

Despite its theoretical success in reducing the required number of global communication rounds, an in-depth structural analysis of the SHAPEFL parser logic and algorithmic implementation reveals severe bottlenecks that limit its scalability, computational efficiency, and real-world applicability in dynamic networks.

The primary bottleneck resides in the computational complexity of the diversity measurement algorithm. To actively shape the data distribution at the edge, the SHAPEFL parser must continuously measure the data diversity among the incoming client updates. The framework utilizes the cosine distance between the high-dimensional model updates to quantify this diversity.⁷ Let Δw_i and Δw_j represent the flattened gradient update vectors received from clients i and j . The cosine distance D_{cos} is computed via the parser as:

$$D_{cos}(\Delta w_i, \Delta w_j) = 1 - \frac{\Delta w_i \cdot \Delta w_j}{\|\Delta w_i\|_2 \|\Delta w_j\|_2}$$

For modern deep neural networks or complex models where the parameter dimension d frequently exceeds tens of millions, calculating the pairwise cosine distance for K clients at the edge aggregator requires $\mathcal{O}(K^2 \cdot d)$ floating-point operations per communication round. This computational overhead is extraordinarily expensive for resource-constrained edge servers, creating a processing bottleneck that offsets the gains achieved in communication efficiency.⁷ The parser must iterate through every combination of client updates, calculate the dot product, compute the individual L_2 norms, and evaluate the distance, which consumes vast amounts of memory bandwidth and CPU cycles at the edge layer.

To mitigate this severe $\mathcal{O}(K^2 \cdot d)$ complexity, the SHAPEFL implementation often defaults to

a highly restrictive workaround: manual layer selection. The parser is configured to extract only specific layers of the neural network to compute the cosine distance, ignoring the rest of the model.⁷ This introduces a manual, human-in-the-loop dependency that breaks the automation pipeline of the distributed learning system. Furthermore, it assumes that the selected layers are universally representative of the data diversity across all clients, an assumption that frequently fails in complex, non-convex optimization landscapes.

An additional structural flaw in the SHAPEFL logic is its reliance on offline pre-training. The initial edge server selection and the baseline data shaping matrices rely heavily on offline pre-training phases to approximate the data distribution before the actual federated training begins.⁷ In authentic distributed environments where client availability is highly stochastic, device dropouts are frequent, and local data distributions are dynamic and non-stationary, offline pre-training is fundamentally unsuited. It leads to sub-optimal node-edge associations that decay in effectiveness as the real-time data distribution drifts away from the pre-trained approximation.⁷ Finally, SHAPEFL operates under the assumption of semi-honest nodes. It lacks robust cryptographic or statistical mechanisms to verify the integrity of the updates, rendering the entire data-shaping paradigm highly susceptible to adversarial nodes that can easily manipulate the cosine distance metrics through strategic gradient poisoning.¹²

Structural and Theoretical Evaluation of the ROSE Implementation

The ROSE implementation approaches distributed optimization from a fundamentally different perspective, prioritizing security and privacy above all other metrics. Motivated by the severe vulnerabilities of standard aggregation algorithms to malicious actors, data leakage, and unauthorized model reconstruction, ROSE alters the foundational mathematics of the parameter exchange.²

Security Mechanisms and Model Update Based Aggregation

In standard federated architectures, the central server aggregates the raw model weights uploaded by participating clients. The theoretical logic behind ROSE recognizes that raw model sharing leaves the entire system vulnerable to membership inference attacks and Byzantine poisoning. In a Byzantine attack, a malicious client might upload a deliberately corrupted vector, such as $w_{malicious} = -\lambda w_{true}$, aiming to destroy the convergence trajectory of the global model.²

To counter this, the ROSE framework implements a Model Update Based algorithm. Instead of transmitting absolute model weights, the parser logic dictates that clients transmit only the localized gradient updates Δw_i^t . The central server then applies a statistically robust aggregation estimator to these updates. If a compromised client attempts an additive noise attack, adding a large random vector ϵ to their update, or a sign-flipping attack, reversing the

direction of their gradient, the MUB estimator identifies the statistical deviation of these vectors from the mean distribution of the benign majority.² The parser dynamically filters, clips, or completely discards these anomalous updates, preserving the integrity of the global model.

Additionally, the ROSE implementation employs an individual client model initialization method. By assigning distinct, mathematically masked initial state vectors to individual clients at the commencement of training, the framework effectively obfuscates the true trajectory of the local gradients. This structural defense prevents the central server, edge nodes, or passive eavesdroppers from reverse-engineering the client's local dataset via membership inference attacks, thereby ensuring strict privacy compliance.²

Algorithmic Bottlenecks in the ROSE Parser Logic

While the mathematical proofs underlying the ROSE methodology confirm its theoretical robustness against Byzantine faults and privacy breaches, an analysis of the source code logic and implementation parameters reveals significant constraints regarding communication efficiency and convergence speed.

The most prominent bottleneck is the reliance on a single-tier, flat network topology. Unlike the hierarchical structure of SHAPEFL, the standard ROSE implementation routes all client communication directly to a singular central parameter server.² In large-scale cross-device deployments involving thousands of IoT devices or mobile endpoints, this flat architecture creates a massive ingress communication bottleneck. The central server must receive, verify, and aggregate every individual dense update vector using the computationally heavy MUB estimator. As the number of participating clients increases, the per-round latency scales linearly, severely degrading the overall communication efficiency and wall-clock training time of the system.

Furthermore, the ROSE algorithm experiences severe convergence degradation under highly non-IID data regimes. The MUB estimator functions by analyzing the variance and geometric distance among client updates to isolate anomalies. However, in highly heterogeneous scenarios, benign clients possessing statistically rare but legitimate data distributions produce updates that naturally deviate significantly from the global mean. The robust estimator logic within the ROSE parser risks misclassifying these highly divergent, benign updates as Byzantine anomalies, thereby down-weighting or discarding them.² This statistical phenomenon inadvertently suppresses the inclusion of diverse, localized data features into the global model. Consequently, the algorithm requires exponentially more communication rounds to converge to an acceptable level of global accuracy, exacerbating the already severe communication bottleneck of the flat topology. The system transmits uncompressed, dense parameter arrays directly across the network, consuming vast amounts of bandwidth without the benefit of the communication-reduction strategies seen in modern quantization or sparsification techniques.³

Deep Comparative Analysis and Bottleneck Synthesis

To synthesize the structural and theoretical analyses of both methodologies, a comparative

evaluation of the inherent bottlenecks in the SHAPEFL framework and the ROSE implementation reveals a fundamental tension in modern distributed optimization. The evaluation relies on the primary parameters of Network Architecture, Client Association, Diversity Measurement, Adversarial Resilience, and Payload Efficiency.

Optimization Parameter	SHAPEFL Framework Logic	ROSE Implementation Logic	Comparative Structural Bottleneck Identification
Network Architecture	Hierarchical (Cloud-Edge-Client topology)	Flat (Single Server-Client topology)	SHAPEFL significantly reduces global communication rounds (κ) by utilizing intermediate edge nodes. ROSE's flat topology creates overwhelming central server bottlenecks and high latency. ³
Client Association Policy	Static, based on offline pre-training	Dynamic, random or full participation	SHAPEFL's reliance on static offline pre-training makes it brittle in dynamic environments. ROSE adapts to changing active sets but suffers from straggler effects and network congestion. ²
Data Diversity Metric	Cosine Distance ($\mathcal{O}(K^2 \cdot d)$ complexity)	Variance Estimation (MUB statistical filter)	SHAPEFL incurs prohibitive computational overhead at the edge for large

			model dimensions. ⁷ ROSE struggles to mathematically differentiate between legitimate non-IID data updates and malicious Byzantine noise. ²
Adversarial Resilience	Extremely Low (Assumes semi-honest participating nodes)	Extremely High (Byzantine fault tolerant design)	SHAPEFL is highly susceptible to sign-flipping and additive noise at the edge level, destroying the diversity metric. ¹² ROSE excels in maintaining privacy and structural security. ⁵
Payload Efficiency	Moderate (Reduced total rounds, uncompressed payloads)	Low (High total rounds, uncompressed dense updates)	Both frameworks transmit uncompressed dense vectors. ROSE is particularly penalized as it requires high total communication rounds under non-IID conditions, consuming massive bandwidth. ³

The critical insight derived from this comparative analysis is that the security mechanisms of the ROSE implementation and the communication efficiency mechanisms of the SHAPEFL framework currently operate in direct algorithmic opposition. The SHAPEFL logic requires the explicit measurement of update diversity to intentionally shape the data distribution at the edge, actively seeking out divergent gradients. Conversely, the ROSE logic seeks to mask and homogenize updates to prevent adversarial manipulation, actively penalizing divergent gradients. Therefore, merely combining the two codebases would result in systemic failure, as the ROSE MUB estimator would systematically discard the highly diverse updates that the

SHAPEFL aggregator works to select.

Proposed Novel Modifications to the ROSE Algorithm

To surpass the performance metrics of the SHAPEFL base framework—specifically regarding convergence speed, communication efficiency, and model accuracy—while retaining the robust cryptographic and statistical security profile of the ROSE implementation, a unified, multi-phasic architectural enhancement is proposed. This architectural evolution, formalized as the Hierarchical-Secure ROSE (HS-ROSE) framework, introduces three novel structural modifications: the integration of a Hierarchical MUB topology, the implementation of Homogeneity-Aware Dynamic Client Selection, and the application of Secure Adaptive Gradient Quantization.

Modification 1: Integration of Hierarchical MUB (H-MUB) Topology

The primary structural limitation of the original ROSE parser is its flat, single-tier architecture, which creates central processing and bandwidth bottlenecks.² The proposed modification transitions the foundational ROSE algorithm to a Hierarchical Federated Learning topology, mapping directly to the physical topology of modern edge computing and telecommunication networks.

In the HS-ROSE architecture, the overall network is physically partitioned into disjoint geographical or logical clusters, each governed by an intermediate Edge Aggregator. The monolithic Model Update Based algorithm is mathematically and logically decoupled into two distinct, sequential operational phases:

1. **Edge-Level Intra-Cluster Aggregation:** The Edge Aggregators act as the primary receivers for the local gradients Δw_i^e generated by their associated clients. The parser logic at the edge server executes a lightweight, localized variant of the MUB estimator. This edge-level estimator is calibrated to filter out extreme, localized Byzantine attacks, such as massive additive noise or complete sign inversion, operating on a smaller subset of total clients. The edge server then computes a verified, intermediate edge-model update ΔW_e .
2. **Cloud-Level Inter-Cluster Aggregation:** The central cloud server receives only the intermediate, aggregated updates ΔW_e from the limited number of Edge Aggregators. Because the edge servers have already filtered the primary local noise and validated the update integrity, the cloud server executes a global, weighted MUB aggregation to form the new global model W_{t+1} .

The theoretical justification for this architectural shift is rooted in distributed systems complexity. By offloading the primary MUB verification and distance calculations to the distributed edge nodes, the computational load on the central server is exponentially reduced

from $\mathcal{O}(N \cdot d)$ to $\mathcal{O}(N_e \cdot d)$, where the number of edge servers N_e is orders of magnitude smaller than the total number of clients N . This topological shift completely resolves the central communication bottleneck of the original ROSE implementation, drastically reducing per-round latency and network congestion, aligning with the core topological advantages of SHAPEFL without adopting its vulnerabilities.³

Modification 2: Homogeneity-Aware Dynamic Client Selection

To address SHAPEFL's reliance on computationally expensive cosine distance metrics and its crippling dependence on offline pre-training⁷, the HS-ROSE framework integrates a Homogeneity-Aware Dynamic Client Selection mechanism. This approach draws upon advanced theoretical concepts from federated pruning and homogeneity-aware aggregation algorithms.⁷

Instead of iterating through the computationally exhaustive process of calculating the cosine similarity of the entire high-dimensional gradient vector (a process with $\mathcal{O}(d)$ complexity per pair), HS-ROSE implements a lightweight Sub-space Homogeneity (SH) scoring logic.

Let $P \in \mathbb{R}^{d \times p}$ be a universal, deterministically generated pseudo-random projection matrix, where the projection dimension p is vastly smaller than the original model dimension d ($p \ll d$). The client-side parser logic is modified such that clients project their high-dimensional gradient updates into this low-dimensional subspace prior to any diversity evaluation:

$$v_i = (\Delta w_i)^T P$$

The edge aggregator then calculates the SH-score using the statistical variance of these low-dimensional projections v_i . Rather than relying on static, pre-trained matrices, the edge server dynamically selects clients for the subsequent communication round based on these real-time SH-scores. The parser logic prioritizes clients whose SH-scores indicate high variance relative to the current edge-aggregate, ensuring that the selected subset of clients provides a diverse, pseudo-IID representation of the global data distribution.

This modification yields profound theoretical benefits. First, computing the diversity metric within the projected subspace mathematically reduces the computational complexity at the edge aggregator from $\mathcal{O}(K^2 \cdot d)$ to $\mathcal{O}(K^2 \cdot p)$. This effectively solves the primary computational bottleneck of the SHAPEFL framework, eliminating the need for manual layer selection and enabling the system to evaluate the entire model structure efficiently.⁷ Second, because the edge aggregator utilizes the SH-score exclusively to select clients—thereby shaping the physical data distribution prior to training—rather than altering the statistical

weights of the MUB estimator itself, HS-ROSE successfully shapes the data distribution for faster convergence without compromising the Byzantine fault tolerance established by the original ROSE logic.² The global model converges with significantly fewer communication

rounds (K) because the strategically selected edge aggregations emulate an optimal IID distribution, bypassing the convergence degradation issue inherent to the baseline ROSE implementation.

Modification 3: Secure Adaptive Gradient Quantization

While the hierarchical topology mathematically reduces the total *number* of global communication rounds, the absolute *size* of the payload transmitted during each round remains a critical bottleneck in the baseline ROSE implementation.¹⁴ To comprehensively exceed SHAPEFL's communication efficiency metrics, HS-ROSE incorporates a secure Adaptive Gradient Quantization module within the parser logic.

Standard quantization techniques map continuous gradient components to discrete intervals, reducing mathematical precision but significantly compressing the total amount of data transmitted across the network (e.g., converting 32-bit floating-point parameters to 8-bit integers).¹⁴ However, applying aggressive, static quantization uniformly across all model layers frequently destroys critical gradient information, leading to severe convergence degradation and instability.¹⁴

The HS-ROSE parser implements a variance-aware, adaptive quantization scheme. During the local update phase, the client-side logic computes the internal variance of the local gradient update, denoted as $\sigma^2(\Delta w_i)$, across the different layers of the neural network. Layers exhibiting high variance are identified as containing critical, highly diverse features necessary for global convergence. The parser dynamically assigns these high-variance layers to be quantized at higher bit-widths (e.g., 8-bit or 16-bit precision) to preserve their informational integrity. Conversely, layers exhibiting low variance are subjected to aggressive compression (e.g., 1-bit or 4-bit precision), maximizing bandwidth savings without sacrificing essential model knowledge.¹⁴

The theoretical justification for integrating this adaptive quantization specifically within the ROSE framework is exceptionally strong. The integration ensures that the absolute communication volume is reduced by significant margins per round, addressing the payload bottleneck directly.¹⁴ Furthermore, the process of quantization inherently introduces structural noise into the transmitted gradients. Because the foundational logic of the ROSE MUB estimator is explicitly designed to be robust against additive noise attacks and statistical perturbations², it naturally absorbs, smooths out, and corrects the quantization error introduced by the compression module. This symbiotic interaction between the adaptive compression logic and the robust security estimator results in a highly accurate global model that communicates much more efficiently than the baseline SHAPEFL architecture, representing a true synthesis of security and performance.

Theoretical Justification for Superiority over SHAPEFL

The integration of these three architectural modifications into the cohesive HS-ROSE framework is projected to yield strictly superior performance metrics across all critical axes of distributed optimization when compared to the baseline SHAPEFL architecture.

Regarding **convergence speed**, by adopting SHAPEFL's core principle of hierarchical data-shaping but executing it through the dynamic, projection-based HA-DCS mechanism, HS-ROSE ensures that the intermediate edge updates continuously approximate an ideal IID distribution. Federated optimization theory rigorously dictates that operating on IID data significantly accelerates the convergence trajectory of Stochastic Gradient Descent algorithms.² By entirely eliminating SHAPEFL's reliance on brittle offline pre-training, HS-ROSE seamlessly adapts to dynamic client data shifts and device dropouts in real-time, maintaining a steep and efficient convergence curve even in highly volatile, non-stationary data environments.

Regarding **communication efficiency**, the combination of the hierarchical topology—which drastically reduces the required number of total communication rounds (K)—and the adaptive gradient quantization module—which drastically reduces the payload size transmitted during each individual round—produces a multiplicative reduction in total system bandwidth consumption. Theoretical models project that HS-ROSE will utilize only a fraction of the total network bandwidth required by the original ROSE implementation, while comprehensively surpassing the efficiency gains achieved by SHAPEFL, which transmits uncompressed, high-dimensional arrays.⁶

Finally, regarding **model accuracy and robustness**, SHAPEFL's foundational assumption of semi-honest nodes leaves the entire framework theoretically vulnerable. A single strategically poisoned client executing a sign-flipping attack can skew the edge aggregator's cosine-distance distribution metric, destroying the accuracy of the global model.¹² HS-ROSE circumvents this vulnerability entirely. By utilizing the Hierarchical MUB estimator at both the edge and cloud levels, the framework actively filters adversarial updates before they can infect the intermediate or global models. This rigorous, multi-tiered security logic guarantees a significantly higher upper bound on final test accuracy in practical, adversarial edge-computing environments compared to the defenseless SHAPEFL framework.²

Research Proposal and Implementation Plan

To rigorously validate the theoretical advantages of the proposed HS-ROSE framework against the baseline implementations of SHAPEFL and standard ROSE, the following structured research and implementation plan is submitted for the execution of the thesis. This plan details the software engineering, simulation, and analytical phases required to generate empirical evidence supporting the proposed architectural enhancements.

Phase 1: Framework Scaffolding and Baseline Recreation (Months 1-3)

The initial phase of the research focuses on establishing a highly controlled, reproducible testing environment and accurately recreating the baseline algorithms for subsequent benchmarking.

The simulated federated network will be deployed utilizing the Flower or PySyft frameworks, which provide robust environments for decentralized machine learning, integrated with PyTorch for the underlying neural network training logic.¹⁶

The original ROSE MUB algorithm will be implemented according to its published mathematical specifications, ensuring the accurate replication of the individual client initialization protocols and the robust variance-based aggregation estimator.² Concurrently, the SHAPEFL hierarchical topology and its associated cosine-distance-based node-edge association algorithms will be coded, including the implementation of the requisite offline pre-training modules to serve as the baseline comparison.⁶ The primary deliverable for this phase is a fully functional benchmarking suite capable of simulating varying degrees of non-IID data distributions (utilizing Dirichlet distribution partitioning) and injecting controlled Byzantine adversaries, specifically targeting sign-flipping and additive noise attack vectors.

Phase 2: Algorithmic Development of the HS-ROSE Enhancements (Months 4-7)

The second phase involves the core software engineering effort: coding and integrating the three novel algorithmic modifications into the baseline ROSE architecture to create the unified HS-ROSE framework.

The central server logic will be modified to support the two-tier cloud-edge aggregation topology, implementing the H-MUB estimator at both the intermediate edge nodes and the central cloud node. Following this structural shift, the random projection logic for the Homogeneity-Aware Dynamic Client Selection mechanism will be developed. The parser logic at the edge aggregators will be updated to compute the Sub-space Homogeneity score, utilizing this metric to dynamically dictate client selection per round without relying on offline pre-training data.⁷ Finally, the variance-aware adaptive gradient quantization module will be integrated into the client-side backward pass functions, ensuring that gradients are compressed prior to transmission based on layer-wise variance thresholds.¹⁴ The completion of this phase will result in the finalized source code for the complete HS-ROSE framework.

Phase 3: Comprehensive Benchmarking and Ablation Studies (Months 8-10)

The third phase is dedicated to generating the empirical data necessary to support the theoretical claims established in this report. This involves running extensive simulations across a variety of hardware constraints and network conditions.

Standardized deep learning models, such as complex Convolutional Neural Networks (e.g.,

ResNet architectures) and recurrent networks, will be trained on established benchmark datasets including CIFAR-10, FEMNIST, and RCV1 to ensure the generalizability of the findings across visual and linguistic domains.¹⁴ The testing suite will configure highly non-IID data partitions across the simulated clients to stress-test the data-shaping capabilities of the HA-DCS module.

To evaluate adversarial resilience, a controlled percentage of the client pool (ranging from 10% to 30%) will be programmed as Byzantine actors, executing coordinated attacks against the aggregators.² During these simulations, the system will record rigorous metrics: the total number of communication rounds required to achieve target test accuracy, the total absolute byte volume transmitted across the network, the computational time elapsed per round at the edge nodes, and the final converged model accuracy. These empirical datasets will facilitate direct, quantitative comparisons between the proposed HS-ROSE framework, standard FedAvg, baseline ROSE, SHAPEFL, and contemporary hierarchical solutions like FedPhD.⁷

Phase 4: Statistical Data Analysis and Thesis Compilation (Months 11-12)

The final phase involves the synthesis of the empirical findings into the concluding academic document.

Rigorous statistical significance testing will be conducted on the collected performance metrics to validate the variance between the frameworks. The empirical results will be systematically mapped back to the theoretical mathematical proofs regarding convergence acceleration and computational complexity reduction established in the early chapters of the thesis. The final document will detail the structural evolution from the isolated paradigms of ROSE and SHAPEFL to the unified, highly optimized HS-ROSE architecture, concluding the research effort and contributing a significant new methodology to the field of secure distributed optimization.

Works cited

1. SHARE: Shaping Data Distribution at Edge for Communication-Efficient Hierarchical Federated Learning | Request PDF - ResearchGate, accessed on April 18, 2026, https://www.researchgate.net/publication/355071748_SHARE_Shaping_Data_Distribution_at_Edge_for_Communication-Efficient_Hierarchical_Federated_Learning
2. A New Implementation of Federated Learning for Privacy and Security Enhancement, accessed on April 18, 2026, <https://par.nsf.gov/biblio/10394978-new-implementation-federated-learning-privacy-security-enhancement>
3. Federated Learning: Navigating the Landscape of Collaborative Intelligence - MDPI, accessed on April 18, 2026, <https://www.mdpi.com/2079-9292/13/23/4744>
4. Federated Continual Learning: Concepts, Challenges, and Solutions - arXiv, accessed on April 18, 2026, <https://arxiv.org/html/2502.07059v1>

5. A New Implementation of Federated Learning for Privacy and Security Enhancement - arXiv, accessed on April 18, 2026, <https://arxiv.org/abs/2208.01826>
6. Hierarchical Federated Learning Framework | PDF | Applied Mathematics - Scribd, accessed on April 18, 2026, <https://www.scribd.com/document/847994399/A-Communication-Efficient-Hierarchical-Federated-Learning-Framework-via-Shaping-Data-Distribution-at-Edge>
7. A Communication-Efficient Hierarchical Federated Learning Framework via Shaping Data Distribution at Edge | Request PDF - ResearchGate, accessed on April 18, 2026, https://www.researchgate.net/publication/378264438_A_Communication-Efficient_Hierarchical_Federated_Learning_Framework_via_Shaping_Data_Distribution_at_Edge
8. [2003.01344] Adaptive Federated Learning With Gradient Compression in Uplink NOMA, accessed on April 18, 2026, <https://arxiv.org/abs/2003.01344>
9. Reactive Orchestration for Hierarchical Federated Learning Under a Communication Cost Budget - ResearchGate, accessed on April 18, 2026, https://www.researchgate.net/publication/386454694_Reactive_Orchestration_for_Hierarchical_Federated_Learning_Under_a_Communication_Cost_Budget
10. A Communication-Efficient Hierarchical Federated Learning Framework via Shaping Data Distribution at Edge - IEEE Computer Society, accessed on April 18, 2026, <https://www.computer.org/csdl/journal/nt/2024/03/10436059/1UvfVM5fzEI>
11. Fed-GTopK: Expediting In-Network Federated Learning by Transmitting Global Top Model Updates | Request PDF - ResearchGate, accessed on April 18, 2026, https://www.researchgate.net/publication/400926627_Fed-GTopK_Expediting_In-Network_Federated_Learning_by_Transmitting_Global_Top_Model_Updates
12. [2409.15067] SHFL: Secure Hierarchical Federated Learning Framework for Edge Networks - arXiv, accessed on April 18, 2026, <https://arxiv.org/abs/2409.15067>
13. A New Implementation of Federated Learning for Privacy and Security Enhancement - DigitalCommons@UNL - University of Nebraska–Lincoln, accessed on April 18, 2026, <https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1721&context=electricaleengineeringfacpub>
14. Studying Imperfect Communication In Distributed Optimization Algorithm - Chalmers ODR, accessed on April 18, 2026, <https://odr.chalmers.se/bitstreams/9fce76e2-e1d0-4f3e-b3fa-18f55bac3009/download>
15. Privacy-Enhanced Federated Learning: A Restrictively Self-Sampled and Data-Perturbed Local Differential Privacy Method - MDPI, accessed on April 18, 2026, <https://www.mdpi.com/2079-9292/11/23/4007>
16. Helmsman: Autonomous Synthesis of Federated Learning Systems via Multi-Agent Collaboration - arXiv, accessed on April 18, 2026, <https://arxiv.org/html/2510.14512v1>
17. Implementation and Evaluation of a Federated Learning Framework on Raspberry PI Platforms for IoT 6G Applications - White Rose Research Online, accessed on April 18, 2026,

- <https://eprints.whiterose.ac.uk/id/eprint/238996/1/futureinternet-15-00358-v2.pdf>
18. Four-Dimension Deep Learning Method for Flower Quality Grading with Depth Information, accessed on April 18, 2026,
<https://www.mdpi.com/2079-9292/10/19/2353>